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1 Abstract

Health is the most important thing for anyone, and existing of a care center near to you is one of the essential things in our lives to keep track of our health from teeth to Pregnancy and Birth. Therefore, we chose the field of care centers in the Kingdom of Saudi Arabia. We used the data on the open data platform, and we started our journey of arranging, organizing, and sorting this data. To make this data more clear to decision-makers in this field. Our division was according to region, then nationality, then the job (nurse, doctor, etc.), and finally according to gender. We also showed the types of clinics and laboratories. The final results were comprehensive for every care center and their facilities in Saudi including the capacity of clinics, different workers, laboratories, and patients' satisfaction with the services and help they get during the recovery process. And it is clear to understand the differences between them to facilitate problem-solving for the managers and decision-makers to if needed.

2 Introduction

A health care center or community health center is part of a network of clinics staffed by a group of general practitioners and nurses to provide health care services to the population of a particular area. It is the place from which most medical consultations and referrals begin. It is also the place from which treatment begins, even when patients are referred to specialists elsewhere for medical care. Primary healthcare centers are important in increasing economic growth and raising production efficiency and service quality. These centers focus on scientific planning and following quality standards based on achieving these goals. They face an absence of activating regulating laws or sufficient funding to implement strategies and plans, which limits the quality of service provided. The aim of the dashboard is to show the numbers of health workers in primary health care centers in all regions of the Kingdom in 4 functional areas (medicine, pharmacy, allied health personnel, nurses) with the possibility of viewing them by gender or nationality or without discrimination, in addition to the visits of the auditors to 6 Clinics for primary health care centers (general clinic, chronic diseases clinic, dental clinic, pregnant women clinic, healthy child clinic, and other clinics) in addition to the number of radiological and laboratory tests. right at the right time.

3 Method

Power BI is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights. The data of the health centers was in the form of a table in the Excel program, but by using Power BI, we were able to show all the data in graphic forms and maps. To make decision-making clear and easy.

- The tool IMAGE is used for representing and describing some jobs or genders.
- The tool TEXT BOX is used to describe or type a name or a purpose.
- The tool SHAPE is used to create a geometrical shape as a small background to display the images or texts more.
- The tool FILTER is used for filtering the wanted regions from the data.

- The tool BOOKMARK is used for capturing the state of a the element in the pages and to link them together.
- The tool BOTTON is used with a bookmark to hide and show some elements on one page, also it's used for a page navigation to navigate between pages easily. Moreover, we benefited from using it as a GIF in the home page.
- The tool GAUGE is used for displaying the patients satisfaction quickly over 3 years of surveys.
- The tool CARD is used for showing the exact number of employees in a specific case or region.
- The tool SLICER is used to give the managers or users the choice between several cities.
- The tool MAP is used to enable businesses to use geographic context to position data or illustrate spatial relationships.
- The tool CLUSTERED COLUMN CHART is used for comparing the numbers between laboratory investigations and radiography patients.
- The tool PIE CHART is used for showing the percentage of the whole clinics or one of them with the exact number of clinics visitors.
- The tool PUBLISH is used to share the dashboard with all the power bi users or a certain team.
- The tool NEW MEASUREMENT is used to creat a new relationship from two or more columns or rows.

4 Result

The interactive dashboard that we designed using Power BI consists of 19 pages. In the design, we focused on making the contents of the dashboard clear, understandable, and tidy. We also focused on making the color tones calm and comfortable for the eye, which also facilitates the extraction of information. In the end, we came up with an interactive dashboard that does not need a Power BI expert to use, so decision makers can view it at any time and get the correct and accurate information at the right time to make important decisions.

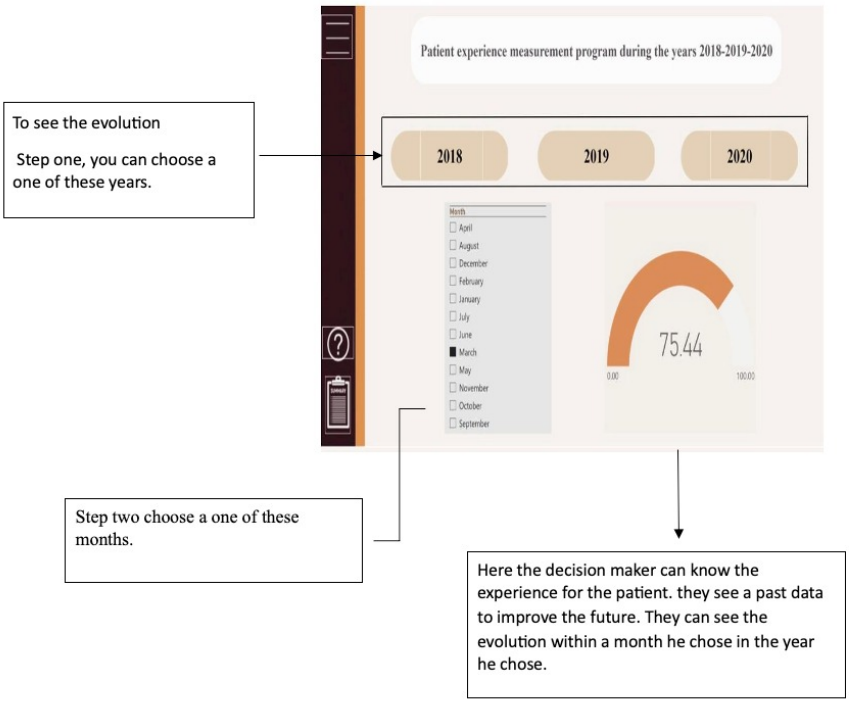

Figure 1: This is an evaluation page of the patient experience in three years. 2018 ,2019, 2020.

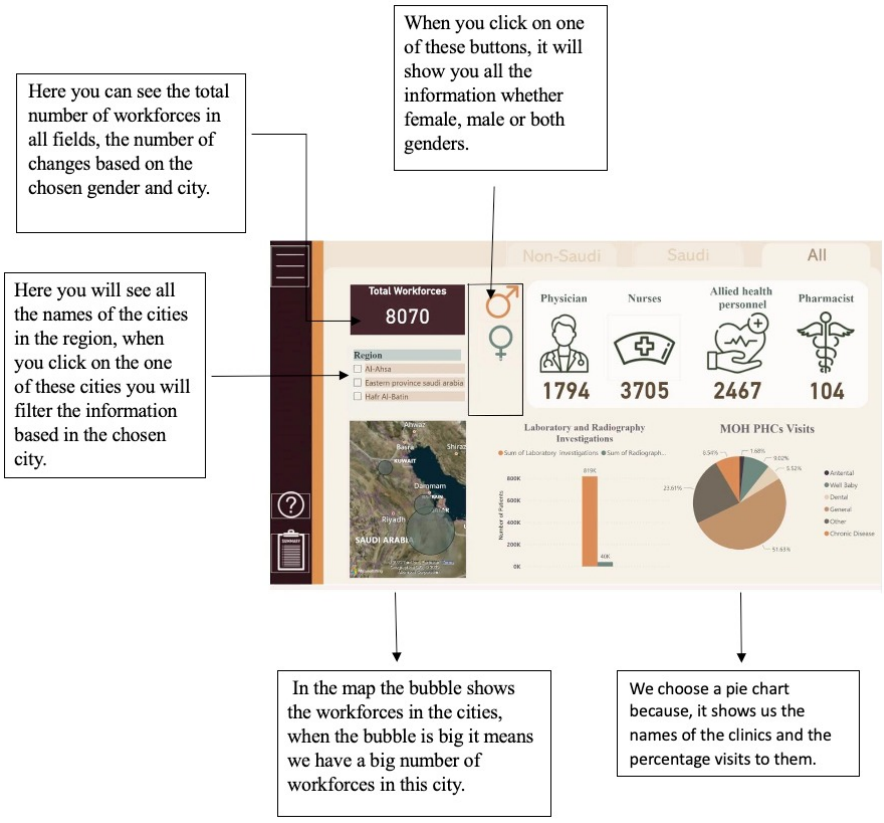

Figure 2: The region dashboard explanation

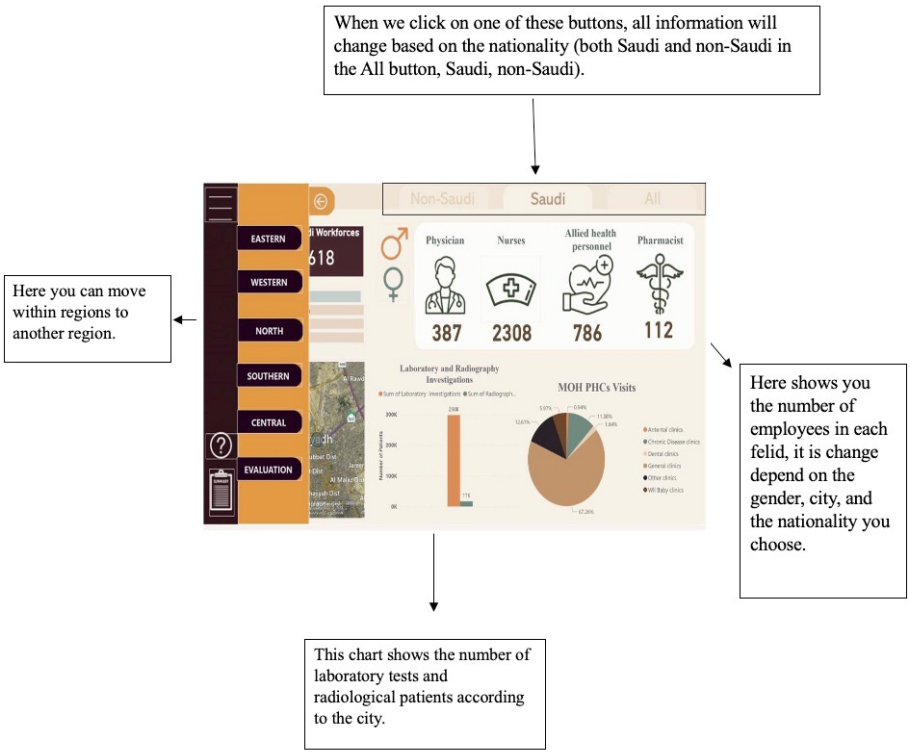

Figure 3: The region dashboard explanation

Figure 4: A summary of the most important points found in the dashboard, to show the summary you must click on one of these buttons, most of them have both text and graph summary

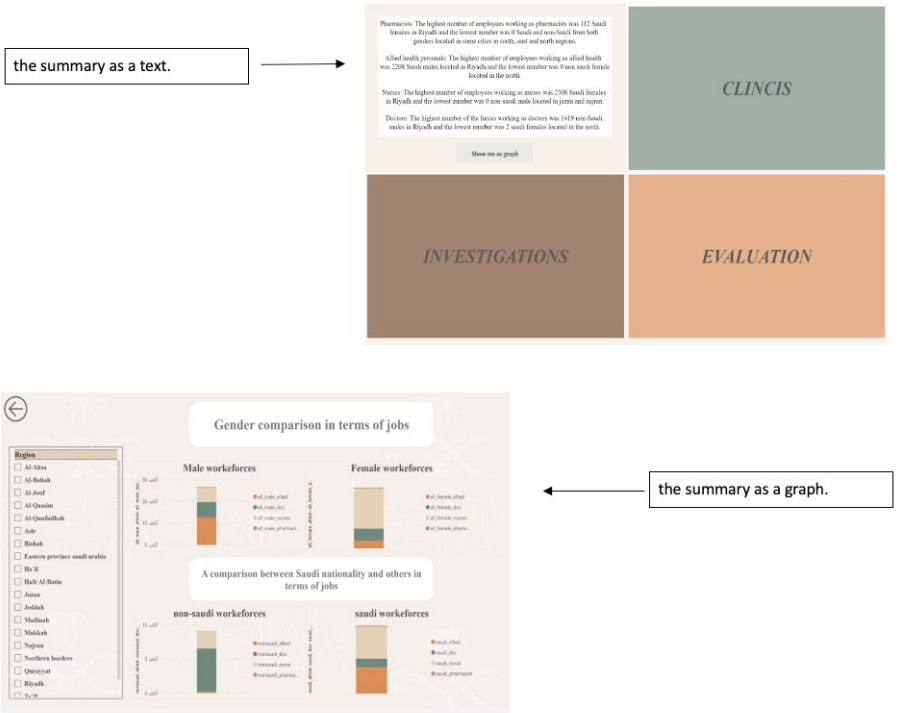

Figure 5: Here is the summary of workforces, you can see a summary as a text or as a graph by clicking on show me as graph button.

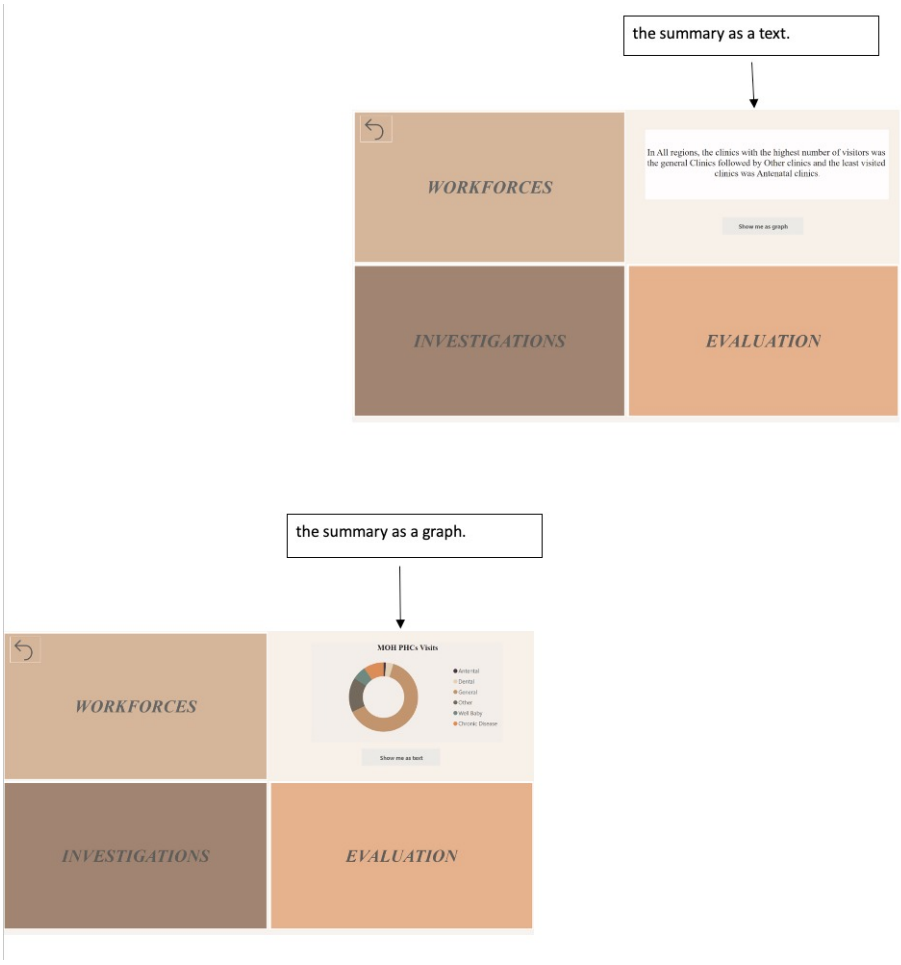

Figure 6: For instance, the clinic summary. you can see a summary as a text or as a graph.

5 Discussion

Firstly, the Eastern Region, which includes the following cities: (Al-Ahsa - Hafar Al-Batin - Dammam, Khobar, and Dhahran). The map showed that the employees of the eastern province cities (Dammam, Khobar, and Dhahran) are higher than the rest of the cities employees, and most of them are Saudi citizens. Also, there was a noticeable discrepancy between male and female employees. For instance, there were 595 male Saudi nurses, while female nurses were 2,798. There is also a decrease in the number of pharmacists compared to other workforces. As for clinics, the general clinics were the most visited clinic, followed by other clinics. Also, there was a convergence between the number of visits to the Chronic Daisies and Well-Baby clinics. Moreover, the radiological examinations were much less visited than the laboratory examinations in the eastern region. Secondly, the western region which includes the following cities:(Al-Qunfudhah, Jeddah, Madinah, Makkah, and Taif). This region was the second-highest in terms of the number of health workforces in it (11.35 employees), and the map showed that most of them are from Madinah and they were Saudi citizens. Also, we noticed the lack of pharmacist employees compared to the rest of the employees. As for the clinics, the most visited clinics in the region were general clinics, followed by other clinics. Furthermore, laboratory tests were the most performed for patients in primary health care centers. Thirdly, The northern region, which included the following cities:(Al-Jawf, Northern Borders, Qurayyat, and Tabuk). It was the lowest region in terms of the number of health workforces in it (3804 employees) and most of them were from Tabuk and Al-Jawf. In addition, there was a noticeable shortage in the number of Saudi pharmacists and doctors in Qurayyat, where there were 2 pharmacists and 20 doctors only. Forth, the Southern region which includes: (Al-Baha, Bisha, Asir, and Najran). The number of employees in PHC centers reached 10.70, and most of them were Saudi citizens. Also, there was a noticeable shortage in the number of pharmacists compared with other workforces number. The last region was the central region, which included the following cities:(Qassim, Hail, and Riyadh). It was the highest region in terms of the number of employees, with 14.74k employees, 11.36 of them were Saudi citizens in Riyadh. However, there was a noticeable shortage of employees in the pharmacy sector, followed by the Saudi doctor sector, as Saudi doctors in it were 470 doctors, while the non-Saudis had 2261 doctors. As for the clinics, the most visited clinics in the region were general clinics, followed by other clinics. Furthermore, laboratory tests were the most performed for patients in PHC centers. Lastly suggestions, PHC centers need more Saudi pharmacists in most regions of the Kingdom and more Saudi doctors in the northern and Southern regions. Also, the focus should be placed on general clinics and their facilities, given that they are the most visited clinics, followed by other, chronic diseases, and well-baby clinics. Furthermore, since most patients that visit PHC centers perform laboratory tests more than radiological examinations, we suggest that more staff should be employed in laboratories and specimen collection sites to speed up the process of obtaining results and improve the client's experience in PHC centers. finally, since both central and southern regions are the highest in terms of the number of health workforces and the number of patient visits, we recommend opening more branches for the PHC centers in the central region, particularly in Riyadh, and in the southern region in Jazan.

6 Conclusion

Finally, we have applied our knowledge of the Power Bi program and designed an interactive dashboard for primary healthcare centers. Since PHC centers provide the most important needs of the individual, family, and society in all stages of life, our primary focus has been to use the best tools to make usage, navigation, and data analysis easy and seamless. Moreover, we have applied the concept of business intelligence, which

is the use of technological techniques such as the dashboard and analyzing previous data to come up with the most important decisions and suggestions for PHC centers. The centers will be pioneers in their field in many aspects, for instance, the patient care and the quality of customer experience, This will result in achieving the goals of primary care centers, as a healthy community and satisfied customers.

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